

Pratham Singla

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EDUCATION

Indian Institute of Technology Roorkee
Bachelor of Technology in Mechanical Engineering

Current CGPA- 9.121/10
August 2023 – Present

EXPERIENCE

- Raeth AI** | *Applied ML Intern* **February 2026 - May 2026**
- Building reinforcement learning environments for finance, designing market simulations and reward structures to train and evaluate trading and decision-making agents.
- Bespoke Labs** | *AI Intern* **October 2025 - January 2026**
- Built a synthetic data generation pipeline simulating realistic user activity to validate a microservices social media platform, and a reward-hacking pipeline exposing vulnerabilities in automated task assessment.
- Lossfunk** | *Research Intern* **December 2024 - September 2025**
- Developed a new benchmark for complex visual reasoning, inspired by ZeroBench, featuring an automated question-generation pipeline to evaluate and train vision-language models on GPT-level reasoning tasks.
- Omega Intelligence** | *Gen AI Intern* **March 2025 - July 2025**
- Built **DE-VTL**, a retrieval framework for RAG: designed an active-learning acquisition function combining diversity, relevance, and uncertainty, and applied LoRA fine-tuning to cut redundancy in retrieved exemplars.
- Boston University** | *Research Intern* **February 2025 - May 2025**
- Investigated LLM self-awareness by analyzing alignment between internal reasoning to outputs, and tested SFT, DPO, GRPO tuned models for bias, risk, reward-hacking, and generalization through targeted experiments.

PUBLICATIONS

- Do Vision–Language Models See or Guess?** | *Under Review at EMNLP'26 (Full paper, First author)* | [Paper](#)
- Built a 540-image phrasing-controlled benchmark with a no-image ablation to isolate VLMs' reliance on textual priors, and showed GRPO post-training of a small VLM yields consistent gains transferring to held-out OOD sets.
- Paradigm Bias in Code Generation: LRLBench** | *Under Review at EMNLP'26 (Short paper)* | [Paper](#)
- Introduced LRLBench (3K problems with verified iterative/recursive references) exposing a 6.9–10.1pp recursion deficit in LLMs; GRPO fine-tuning of Qwen3-4B recovered +4.6pp recursive pass@1 at 1.1pp iterative cost.
- Poker Arena** | *ICML NeXT Game Workshop'26* | [Paper](#)
- Built a no-limit Texas Hold'em tournament platform pairing a three-layer memory architecture with a nine-axis cognitive profile to evaluate seven frontier LLMs beyond scalar leaderboards.
- Reflective Self-Awareness and Reasoning Alignment in LLMs** | *ACL'26 Findings* | [Paper](#)
- Analyzed self-awareness and reasoning alignment in LRMs, such as DeepSeek-R1, across scenarios. Evaluated the impact of SFT, DPO, and GRPO fine-tuning on model behavior, including bias, risk, and reward hacking.
- Give me a hint: Can LLMs take hint to solve math problems?** | *NeurIPS'24 Math-AI Workshop* | [Paper](#) | 🔗 [Code](#)
- Conducted a study on LLMs ability to use hints in mathematical problem-solving on the MATH dataset, benchmarking zero-shot, few-shot, chain-of-thought, and adversarial hinting against traditional prompting.
- Adaptive Urban Planning** | *AAAI'25 AI4UP Workshop* | [Paper](#)
- Developed a Multi-Agent urban planning framework using LLMs and Genetic Algorithms for optimization and regional customization, resulting in significant improvements in livability and accessibility.
- StegaVision: Enhancing Steganography with Attention Mechanism** | *AAAI-25 Student abstract* | [Paper](#) | 🔗 [Code](#)
- Neural image-steganography model integrating channel and spatial attention into an autoencoder.

ACHIEVEMENTS

- 11th / Top 15** – Meta × Huggingface × Pytorch OpenEnv Hackathon, India's largest AI hackathon (70K+ entries).
- Silver Medal** – Inter IIT Tech Meet 14.0, ISRO Geospatial problem statement ([Drishti](#)).
- Winner** – HiLabs AIQuest 2025 Hackathon, IIT Roorkee: data-quality analytics with SLM-based NL querying.

PROJECTS

- RosterIQ: Memory-Augmented Multi-Agent Platform** | *VLG,IITR* | 🔗 [Code](#)
- Multi-agent orchestration (4 agents, 10 tools) with Pydantic-validated tool-calling and a **3-layer memory architecture** (session, episodic, semantic) for persistent reasoning over 60K+ records.
- Voice Agent Autodialer** | *VLG,IITR* | 🔗 [Code](#)
- Developed an AI-powered voice agent auto-dialer system that automates phone surveys by integrating Supabase, Atoms, and Google Sheets for intelligent calling, data collection, and analytics.

SOCIETIES AND EXTRA CURRICULARS

Vision and Language Group, IIT Roorkee | *Secretary* | [Website](#)

Februrary 2024 - Present

ACM, IIT Roorkee | *Project Head* | [Website](#)

March 2024 - Present